

WASP-39b

R-1356

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_210323 · created 2026-07-31T23:06:51+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459296.8768 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.01
Transit depth (Rp/Rs)^2	1.3 +/- 0.24 [%]
Orbital Inclination (inc)	87.26 +/- 0.77
Airmass coefficient 1 (a1)	0.9681 +/- 0.0082
Airmass coefficient 2 (a2)	0.0226 +/- 0.005
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	#1 - [578, 140]
Optimal Method	PSF photometry
Transit Duration (day)	0.108 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.114 $\pm$ 0.01 vs 0.1457 (deviation 2.86 σ)
Duration fit vs published	0.108 d vs 0.12 d (deviation 0.98 σ)
Mid-time O–C	12.9 min
Depth SNR	10.3
Residual scatter	0.84 %

Light curve

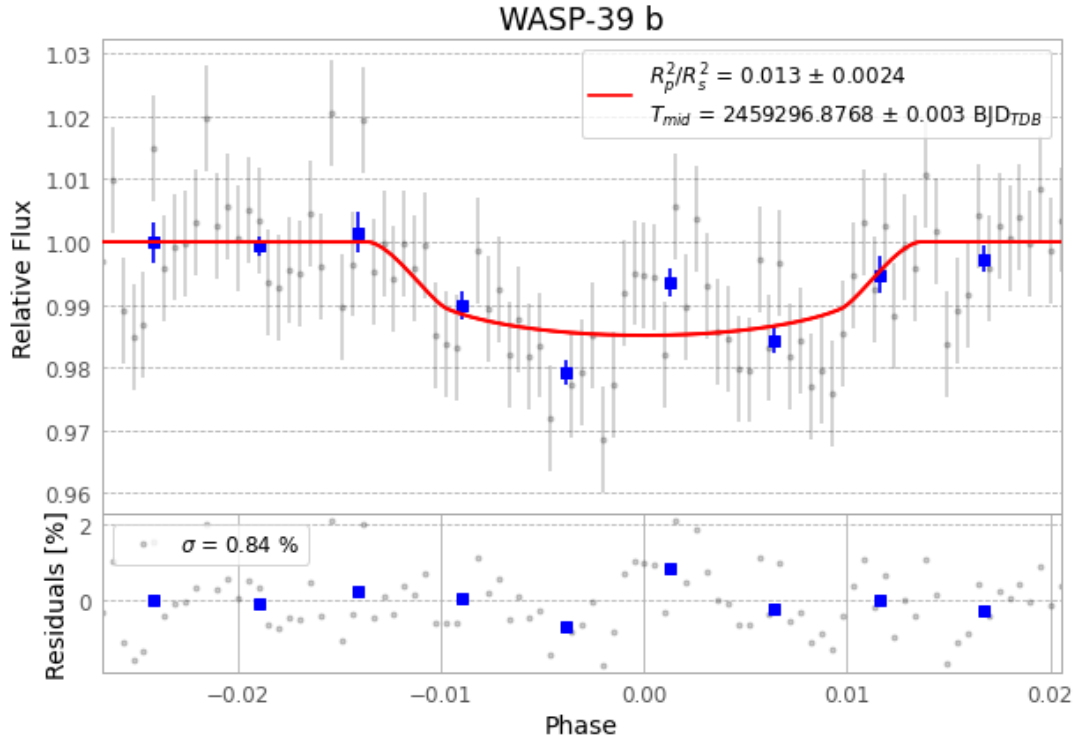


Figure 1 — FinalLightCurve_WASP-39 b_2021-03-22.png (SHA-256 d69d495387b94a13...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 121], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eb5545764be8a705...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), WASP-39210323061812.FITS → WASP-39210323105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-210323.log (a4196772f5c0...), FinalLightCurve_WASP-39 b_2021-03-22.png (d69d495387b9...), FinalLightCurve_WASP-39 b_2021-03-22.csv (5caac47a0978...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:06Z.